

Mahad Aziz

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Technical Skills

- **Languages:** TypeScript, JavaScript, Python, Java, SQL, HTML/CSS, Haskell
- **Frontend:** React, Next.js, D3.js, Tailwind CSS, Adobe CEP/ExtendScript
- **Backend:** Node.js, Express.js, Flask, GraphQL, REST APIs, WebSockets
- **Databases:** PostgreSQL, MongoDB, MySQL
- **Infrastructure:** Docker, Kubernetes, Terraform, AWS, Azure, Jenkins, GitHub Actions
- **Other:** CI/CD, OAuth/JWT (NextAuth.js), Git

Work Experience

Junior Software Engineer

May 2024 – Present

Evertz Microsystems

- Built the Adobe Media Suite from scratch – a React + CEP extension running inside Adobe Premiere, now deployed at multiple broadcast customer sites; delivered sequence/clip import via Adobe Media Encoder with .epr preset workflows, metamarks-to-clip generation, and full clip/sequence management
- Maintain a 5-service broadcast multiviewer design tool now rolling out to customers, shipping 74 merged PRs across it and adjacent Evertz codebases; implemented runtime display-count resizing, full-window widget geometry preserving video layouts, and support for 30 new device layout configurations
- Drove cross-stack reliability work across frontend, server, and the WAMP messaging layer: resolved race conditions in concurrent layout updates, eliminated orphaned canvas state on system initialization, and refactored live-update notification handling
- Optimized device discovery in the management UI to fetch only the 3 relevant device types out of 15 rather than the full device list, cutting irrelevant entries from the clients-grid response by ~80% on typical customer deployments

Student Design Engineer – Software

May 2022 – Apr 2024

Evertz Microsystems

- Built drag-and-drop file uploads for the storage and canvas apps with RESTful endpoints for secure media transfer, plus a remount control for recovering downed mounts and a retry/timeout flow for file-browser failures
- Reworked the device-management UI: replaced the usage column with a filterable jobs view, drove device types from live state instead of hardcoded presets, added client context view/switch, and guarded against mass-rebooting gateways
- Triageed and resolved 150+ feature requests and bugs across React, TypeScript, and Flask; authored technical documentation and mentored two interns through code reviews and pairing sessions

Projects

Collabspace

Demo Code

- Built a real-time collaborative document editor with Next.js 15, React 19, Tiptap v3, and Yjs CRDT in TypeScript strict mode, gated by a custom Node.js WebSocket sync server using NextAuth v5 JWTs and per-document ACLs
- Persisted Yjs state to Postgres on a 10s idle debounce to decouple DB writes from keystroke rate; implemented live cursors and presence via Yjs awareness and Tiptap CollaborationCaret with deterministic userId-to-HSL coloring and multi-tab dedup
- Provisioned AWS infrastructure with Terraform (VPC, ECR, EC2/SSM, IAM, Route53) and a 3-stage GitHub Actions pipeline using OIDC for keyless deploys, shipping 2 containers behind a Caddy TLS/WebSocket proxy

Journey AI

Demo Code

- Built a full-stack data-storytelling assistant (Next.js, Flask, MongoDB) with a pandas/SciPy engine that auto-generates natural-language insights across six lenses — outliers, trends, skew, rankings, Pearson correlations, and descriptive stats over any uploaded dataset
- Integrated Falcon-7B-Instruct (Hugging Face Inference API) to turn selected insights into data-grounded hypotheses, using a rolling 10-entry context window and tuned generation params to curb output drift
- Built a CSV/XLSX/JSON ingestion pipeline (pandas → MongoDB) with automatic column-type inference, powering interactive D3.js bar, line, and scatter charts rendered from each insight's data

Education

McMaster University, Hamilton, ON

Honors Bachelor of Applied Science, Computer Science

GPA: 3.9/4.0